

AY2627

Introduction 簡介

HHS4185 Common Rehabilitation Conditions

HHS4185J 復康科常見疾病

Lecturer 講師

HHS4185 : Ms. Fion Yeung

HHS4185J : Ms. Shan Li

Module Intended Learning Outcomes(MILO)

單元預期學習成果

1. identify and describe the common orthopaedic, medical and neurological conditions in clinical settings.
2. justify the appropriate role of rehabilitation for these common conditions and suggest ways with reference to the clients' conditions to assist therapists to conduct relevant rehabilitation services.

1. 識別並描述臨床環境中常見的骨科、內科和神經系統疾病。
2. 論證復康於這些常見病症的角色，並按服務對象的病況提出建議，以協助治療師進行相關的復康服務。

Assessment 評核

Moodle

- Timetable 時間表
- ❖ Continuous assessment 持續評估 (CA) 50%
 - Three CA Assignments (Individual) 三份 CA 功課 (個人)
 - CA1 10%
 - CA2 10%
 - CA3 10%
 - One CA proposal (Group) 一份CA 計劃書 (小組)
 - CA 4 20%
- ❖ End of module assessment-Project 末期評估 (EA) 50%
 - One EA presentation (Group) 一份EA匯報 (小組)
 - EA1 PPT Group Presentation EA1 PPT 小組匯報 20%
 - One leaflet Production(Group) 製作一份宣傳單張 (小組)
 - EA2 Leaflet EA2 單張10%
 - One Written Test (Individual) 一次筆試 (個人)
 - EA Written Test EA 筆試 20%

Grouping for CA proposal homework & EA presentation

CA 計劃書 & EA 匯報的分組

- Form groups of 6-8 students

HHS4185 Common Rehabilitation Conditions

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Workshop 工作坊

WS1

Introduction to Rehabilitation Equipment (1)

復健器材簡介 (1)

Lecturer 講師

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Agenda 議程

1. Able to recognize & operate common equipment in rehabilitation settings
 - monitoring functions
 - mobility aids
 - body support
2. Able to check, repair & replace those equipment

1. 能夠識別和操作復健環境中的常用設備
 - 監控功能
 - 助行器
 - 身體支撐
2. 能夠檢查、修理和更換這些設備

Equipment in Rehabilitation Settings

復健環境中的設備

1. **Monitoring** – e.g. digital blood pressure monitor, oximeter
 2. **Mobility aids** – e.g. walking aids, wheelchair
 3. **Body support** – e.g. splints & brace
 4. **Aids for activities of daily living (ADL)** – eg Commode, Ripple bed
1. 監測-例如電子血壓計、血氧儀
 2. 助行工具 - 例如助行器、輪椅
 3. 身體支撐 肢體支撐 - 例如夾板和護具
 4. 日常生活活動輔助工具（**ADL**）
- 例如馬桶、氣墊床

Equipment for Monitoring 監控設備

Vital signs measurement 生命體徵測量

blood pressure (BP), pulse (beat/min)

血壓（BP）、脈搏（心跳/分鐘）



Digital Blood Pressure Monitor

手臂式電子血壓計



Digital Blood Pressure Monitor Wrist Cuff
手腕式電子血壓計

Vital signs measurement 量度維生指標

Blood pressure (BP) & pulse (bpm) 血壓（BP）和脈搏（bpm）



Blood Pressure Categories



BLOOD PRESSURE CATEGORY	SYSTOLIC mm Hg (upper number)		DIASTOLIC mm Hg (lower number)
NORMAL	LESS THAN 120	and	LESS THAN 80
ELEVATED	120-129	and	LESS THAN 80
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 1	130-139	or	80-89
HIGH BLOOD PRESSURE (HYPERTENSION) STAGE 2	140 OR HIGHER	or	90 OR HIGHER
HYPERTENSIVE CRISIS (consult your doctor immediately)	HIGHER THAN 180	and/or	HIGHER THAN 120

American Heart Association, 2022

血壓類型



血壓類別	收縮壓毫米汞柱 (mm Hg) (在上面的數值)		舒張壓毫米汞柱 (mm Hg) (在下面的數值)
正常	不到120	及	不到80
偏高	120-129	及	不到80
高血壓，第一期	130-139	或	80-89
高血壓，第二期	140或更高	或	90或更高
高血壓危險期 (請立即諮詢您的醫師)	高於180	及/或	高於120

American Heart Association, 2022

Vital signs measurement- SpO2

量度維生指標

Pulse Oximeter

Oxygen Saturation in
blood- SpO2

Normal level: **95% to
100%(RA)**

<95%: Hypoxemia, low
oxygen level in tissues &
organs (Hypoxia)

S/S: shortness of breath
(SOB), headache,
dizziness, fatigue, bluish
tint to skin / lips,
increase HR



脈搏血氧儀
血液中的氧飽和度 - SpO₂

正常水平：**95% 至 100%** (RA - Room
Air 室內空氣)

<95%：低氧血症、組織器官含氧量低
(缺氧)

S/S：呼吸急促 (SOB)、頭痛、頭暈、
疲倦、皮膚嘴唇變藍、心跳加速

Mobility Aids 助行器

Purpose

devices in **assisting individuals** with **mobility impairment** on **walking & other daily activities**. They could have **greater independence, better balance & safety**. Besides, **Fall prevention** would be enhanced.

Target groups

- Individuals with **injury /disability** that **affect mobility**
- Elderly with **increased fall risk**

eg Walking aids & wheelchair

目的

幫助行動不便的人行走和進行其他日常活動。他們可以擁有**更大的獨立性、更好的平衡性和安全性**。此外，還將加強**預防跌倒**。

對象

- **影響行動能力的受傷/殘疾人士**
- **跌倒風險增加的老年人**

例如助行器和輪椅

Factors affecting Choice of Walking Aids

影響助行器選擇的因素

1. Ability of individuals

- a. **Physical status: muscle strength & endurance, balance /stability during transfer & gait**
- b. **Gait pattern & weight bearing status-**
non weight bearing (NWB),
touch down walking (TDW)
partial weight bearing (PWB),
full weight bearing (FWB)
- c. **Cognitive status** : understand & manage on proper usage
- d. **Pain during transfer & mobility**

1. 個人能力

- a. 身體狀況：轉移和步態過程中的肌力、肌耐力、平衡/穩定性
- b. 步態模式和負重狀態 -
非負重（NWB）、
著地行走（TDW）、
部分負重（PWB）、
全負重（FWB）
- c. 認知狀態：了解和處理正確的使用
- d. 轉移和活動過程中的疼痛

Factors affecting Choice of Walking Aids

影響助行器選擇的因素

2. Level of Assistance

3. Environmental

- landscape: level ground, slope & stairs
- distance travelled usually

2. 幫助程度

3. 環境

- 平地、斜坡和樓梯
- 通常移動的距離



Walking Aids 助行器

One side 單邊：Stick 拐杖/士的

- Least assistance for weight bearing

- 最少的負重協助



Walking Aids 助行器

One side 單邊: **Quadripod** 四腳叉

- **Small based Quadripod** 小四腳叉
 - **Large based Quadripod** 大四腳叉
- Better support than stick with larger base, commonly use for stroke patients
 - 比拐杖提供較多的負重協助, 較常用於中風病人



Small based Quadripod
小四腳叉

Large based
Quadripod
大四腳叉

Walking Aids 助行器

Two side 雙邊- Elbow Crutches 肘拐杖

- for non-weight, partial weight, TDW, full-weight bearing stage
 - usually for adults/ youngsters with orthopedic conditions
 - those with stronger upper limb & core stability
-
- 適用於非負重、部分負重、**TDW**、全重量承載級
 - 通常適用於患有骨科疾病的成人/青少年
 - 上肢較強、核心穩定性較強者



Walking Aids 助行器

Two sides 雙邊 -

Walking Frame 助行架

Rollator 助行輪架 (有輪助行架)

- fixed/ foldable
- 固定/可折疊
- usually for elderly & those with weaker physical, poorer balance conditions
- 通常適用於老年人和身體較弱、平衡狀況較差的人
- for non-weight bearing NWB, partial weight bearing PWB, Touch Down-weight TDW, full-weight bearing FWB stage
- 用於非負重 NWB、部分負重 PWB、觸地負重TDW、全負重 FWB



Walking Frame
助行架



Rollator 助行輪架
(有輪助行架)

Checking & Maintenance walking aids

檢查和保養助行器



1. Appearance

- Loosen parts? Any rusting? damaged handle? deformed shaft? damaged rubber ferrule ?
- any **repair/ replacement** needed ?

2. Proper usage of walking aids

One side: placed at **GOOD side**

Height:

handle at **greater trochanter level**,
elbow slightly bend at **20 to 30 °**

1. 外形

- 鬆動零件？有生鏽嗎？手柄損壞？變形的軸？橡膠套圈損壞？
- 需要維修/更換嗎？

2. 正確使用助行器

一側：放置在好側

高度：

手柄在股骨大轉子水平, 肘部約 20 至 30° 稍微彎曲

Checking & Maintenance walking aids

檢查和保養助行器

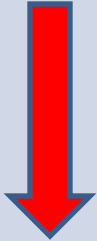
3. Educate clients/patients on means for maintenance & safety

- avoid wet the walking aids to minimize rusting
- regular checking of wear & tear on aids, any loosen parts/crack
- regular replacement of rubber ferrule/ stopper
- slow down when walking on uneven/ slippery floor
- for walking frame, avoid load carrying

3. 對患者進行保養方法和安全意識的教育

- 避免弄濕助行器，以盡量減少生鏽
- 定期檢查輔助工具的磨損情況，任何鬆動的零件/裂縫
- 定期更換橡膠套圈/塞子
- 在不平坦/濕滑的地板上行走時放慢速度
- 對於行走架，避免承載重量

Summary of walking aids 助行器總結

Walking Aids 助行器	Support & stability from walking aids 助行器的支撐和穩定性	Weightbearing Status 負重狀態	Stairs walking 走樓梯
Walking stick 士的	Least 最少  Most 最多 ** Good control/ Assistance 良好的控制/輔助	FWB Only 只有全負重	√
Quadripod 四腳叉		FWB Only 只有全負重	√
Elbow crutches 肘拐杖		NWB 非負重, TDW 觸地 負重, PWB 部份負重, FWB 全負重	√
Frame 助行架		NWB 非負重, TDW 觸地 負重, PWB 部份負重, FWB 全負重	×
Rollator 助行輪架		NWB 非負重, TDW 觸地 負重, PWB 部份負重, FWB 全負重	×

Manual Wheelchair 手動輪椅

Self-propelled wheelchair

自推式輪椅

Youngsters/ adults with

- **Better trunk stability & control**
- Considerable strength & endurance for **upper limb**
- **Sound cognitive**

青少年/成人

- 更好的身體穩定性和控制力
- 上肢相當大的力量和耐力
- 健全的認知



Transit wheelchair

助推式輪椅

Elderly
/Individuals with

- **Weak trunk stability & control**
- +/- **impaired cognitive**
- Need **assistant** to control

長者/有以下情況的人

- 身體穩定性和控制力較弱
- +/- 認知障礙
- 需要助手來控制



Manual Wheelchair 手動輪椅

Inclined wheelchair

傾斜輪椅



- Elderly /individuals with **WEAK physical +/- POOR cognitive status**
- **Weak** head, trunk & lower limb control
- Must with **assistant** to control
- Provide extra support & stability
- Easier manage on **postural hypotension**
- **Heavier**
- 老年人/身體虛弱+/-認知狀況不佳的人
- 頭部、軀幹和下肢控制無力
- 必須有幫助才能控制
- 提供額外的支撐和穩定性
- 更容易管理體位性低血壓
- 較重

Electrical Wheelchair 電動輪椅

Individuals with **Weak physical**
& **sound cognitive**

Battery charging+ Energy
conservation

Independent & efficient
mobility

身體體格較弱，但有健全認
知能力的人

電池充電+節省體能

獨立高效的移動性



Reference

港島東健康資源網 (2022). 助行器使用方法 Use of Walking Aids

<https://www.healthyhkec.org/healthcare/use-of-walking-aids/walking-frame-1/>

醫院管理局 - 助行器的使用 <https://www3.ha.org.hk> > ntwc > pep

醫院管理局- 智友站- 紓緩治療平台-助行器及輪椅應

<https://www21.ha.org.hk/smartpatient/PalliativeCare/zh-HK/CaringTips/Use-of-Walking-Aids-and-Wheelchair/>